

**Department of Finance
Olin Business School
Washington University in St. Louis
Research Methods in Finance**

Stata Group Assignment 3 - Group 10

1. Download industry-level data for the dividend yield or dividend payout ratio for the United States, Europe, Japan, or Emerging Markets from the current data page on Aswath Damodaran's website. Also download industry-level data for the debt-to-capital ratio, market beta, and expected growth in earnings per share (EPS) over the next five years for the same country or country group.

We selected **United States industries** and **dividend yield**. The four source workbooks are dated January 5, 2026 and were downloaded on September 29, 2026. Industries are matched by name, with each industry receiving equal weight.

- Dividend yield: total dividends / market capitalization of equity (divfund.xls).
- Debt-to-capital: market debt-to-capital, adjusted for leases (dbtfund.xls).
- Market beta: levered Beta column (betas.xls).
- Expected EPS growth: expected annual growth over the next five years (pedata.xls).

The matched data contain 94 industries. We exclude the two aggregate Total Market rows and use a common sample of **91 industries** for all regressions. Chemical (Diversified), Real Estate (General/Diversified), and Reinsurance lack EPS-growth forecasts and are omitted. Yield, debt-to-capital, and growth are measured in percentage points; beta is unitless. No outliers are removed.

2. Estimate a linear regression with the dividend yield or dividend payout ratio (debt-to-capital ratio) serving as the dependent (explanatory) variable. In addition, estimate quadratic and cubic regression specifications. Do you find evidence of nonlinearities in the relationship between the dividend yield or dividend payout ratio and the debt-to-capital ratio? Create a scatterplot for the dependent and explanatory variables that includes the fitted linear, quadratic, and cubic regressions.

Question 2 (continued)

Let Y be dividend yield (%) and D be debt-to-capital (%). The fitted OLS equations are:

Linear: $Y = 0.91860 + 0.02636D$

Quadratic: $Y = -0.17621 + 0.11728D - 0.001359D^2$

Cubic: $Y = 0.36860 + 0.04223D + 0.001153D^2 - 0.00002251D^3$

Specification	N	R-squared	Adjusted R-squared
Linear	91	0.1024	0.0923
Quadratic	91	0.2281	0.2106
Cubic	91	0.2435	0.2174

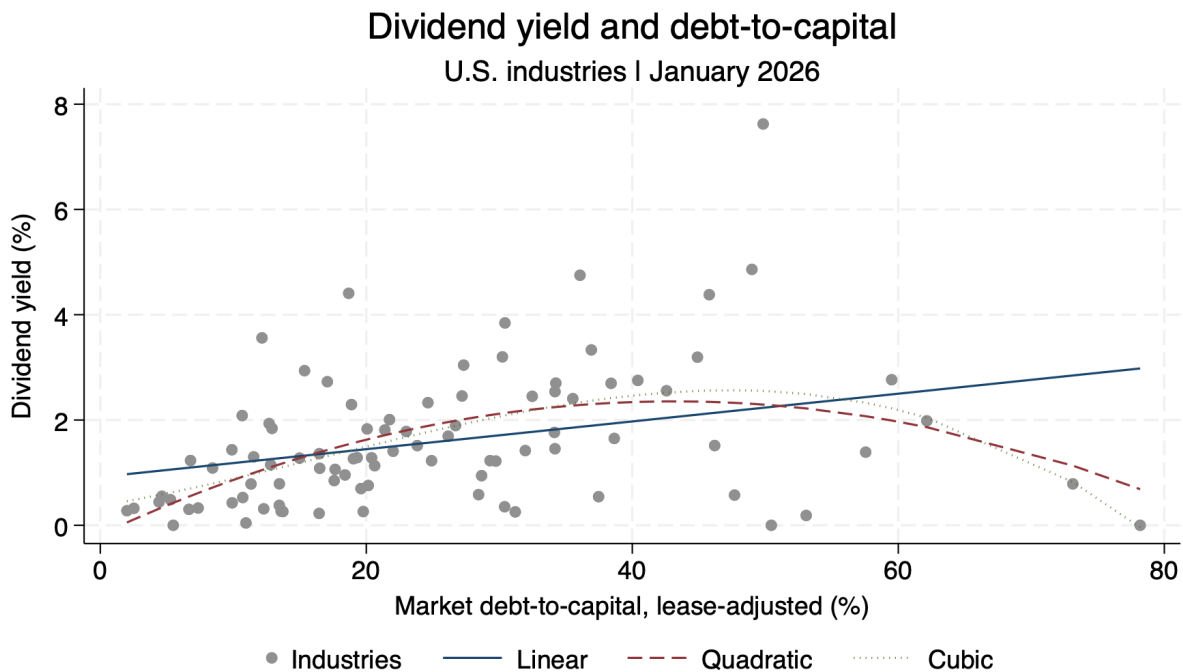


Figure 1. Dividend yield against lease-adjusted market debt-to-capital for 91 U.S. industries, with fitted linear, quadratic, and cubic OLS regressions. Source: Aswath Damodaran, January 2026 industry data.

Question 2 (continued)

There is evidence of **nonlinearity**. In the quadratic model, the squared-debt coefficient is negative and statistically significant ($t = -3.7862$; $p = 0.000279$). Testing the squared term gives $F(1, 88) = 14.34$. R-squared increases from 0.1024 in the linear model to 0.2281 in the quadratic model.

The quadratic fit has an inverted-U shape: dividend yield rises with debt-to-capital up to roughly **43.15%**, then declines. This turning point is computed as $-0.117283 / (2 \times -0.00135898)$. It describes the fitted cross-sectional relationship; it is not an optimal debt ratio or evidence that changing leverage causes dividend yield to change.

The cubic model raises R-squared only modestly, to 0.2435. Its cubic term is not significant ($t = -1.3299$; $p = 0.1870$; $F(1, 87) = 1.77$). A joint test of the squared and cubic terms rejects a purely linear relationship ($F(2, 87) = 8.11$; $p = 0.0006$), but there is no strong evidence that the cubic term adds to the quadratic specification. The quadratic model therefore provides the more parsimonious description of the curvature.

All tests use conventional OLS standard errors. Polynomial predictions are interpreted only within the observed debt-to-capital range; the sparse high-debt observations can influence the fitted curvature.

3. Estimate a multiple linear regression with the dividend yield or dividend payout ratio serving as the dependent variable and the debt-to-capital ratio, market beta, and expected earnings growth serving as the explanatory variables. Report the results (coefficient estimates, standard errors, t-statistics, R-squared statistic) in a table. Discuss the results.

We estimate $Y = a + b_1D + b_2\text{Beta} + b_3\text{Growth} + \text{error}$ using the same 91 industries. The following table reports conventional OLS standard errors.

Variable	Coefficient	Standard error	t-statistic	p-value
Debt-to-capital (%)	0.016198	0.008382	1.9325	0.056547
Market beta	-1.752756	0.494534	-3.5443	0.000636
Expected EPS growth (%)	0.006039	0.008701	0.6941	0.489495
Constant	2.706858	0.577292	4.6889	0.000010

N = 91; R-squared = 0.2157; adjusted R-squared = 0.1887. The overall regression is significant: $F(3, 87) = 7.98$, $p = 0.0001$.

Holding beta and expected growth constant, a 10-percentage-point increase in debt-to-capital is associated with a **0.1620-percentage-point increase** in dividend yield. This coefficient is significant at 10% but not 5% ($p = 0.0565$). Its size and significance are lower than in the simple linear regression.

Market beta is negatively associated with dividend yield. A one-unit increase in beta is associated with a **1.7528-percentage-point decrease** in yield, holding the other variables constant ($p = 0.000636$). This is consistent with higher market-risk industries tending to pay lower dividend yields, although the regression does not establish a causal explanation.

The expected EPS-growth coefficient is small and positive: a 10-percentage-point increase in expected growth is associated with a 0.0604-percentage-point increase in yield. It is not statistically significant ($p = 0.4895$), so the sample provides no clear evidence of a conditional growth-yield relationship.

The model explains about 21.6% of the cross-industry variation in dividend yield. Most variation remains unexplained. These are industry-level associations, not firm-level causal estimates.

4. Provide the Stata code in an appendix.

Appendix: Stata code

The complete executed do-file follows. Run it from the week-04 folder. The original XLS workbooks must remain in the data subfolder.

```
* Stata Group Assignment 3 - Group 10
* Run from the week-04 folder. Original January 2026 XLS files are preserved.
version 19.0
clear all
set more off
cd "."
capture log close
log using "output/assignment3.log", text replace

* 1. U.S. industry data: dividend yield, market debt/capital, beta, EPS growth.
tempfile dividends debt betas
import excel "data/divfund.xls", sheet("Industry Averages") cellrange(A9) clear
keep A F
rename (A F) (industry dividend_yield)
replace industry = strtrim(industry)
drop if industry == "" | strpos(industry, "Total Market") == 1
isid industry
save `dividends'
import excel "data/dbtfund.xls", sheet("Industry Averages") cellrange(A9) clear
keep A F
rename (A F) (industry debt_capital)
replace industry = strtrim(industry)
drop if industry == "" | strpos(industry, "Total Market") == 1
isid industry
save `debt'
import excel "data/betas.xls", sheet("Industry Averages") cellrange(A11) clear
keep A C
rename (A C) (industry market_beta)
replace industry = strtrim(industry)
drop if industry == "" | strpos(industry, "Total Market") == 1
isid industry
save `betas'
import excel "data/pedata.xls", sheet("Industry Averages") cellrange(A9) clear
keep A I
rename (A I) (industry eps_growth)
replace industry = strtrim(industry)
drop if industry == "" | strpos(industry, "Total Market") == 1
isid industry
merge 1:1 industry using `dividends', assert(match) nogen
merge 1:1 industry using `debt', assert(match) nogen
merge 1:1 industry using `betas', assert(match) nogen
foreach v in dividend_yield debt_capital market_beta eps_growth {
    capture confirm string variable `v'
    if !_rc destring `v', replace ignore("NA")
}
}
```

```

* Convert decimal ratios to percentage points; beta remains unitless.
foreach v in dividend_yield debt_capital eps_growth {
    replace `v' = 100 * `v'
}
label variable dividend_yield "Dividend yield (%)"
label variable debt_capital "Market debt/capital, lease-adjusted (%)"
label variable market_beta "Levered market beta"
label variable eps_growth "Expected annual EPS growth, next 5 years (%)"
misstable summarize dividend_yield debt_capital market_beta eps_growth
count
count if missing(dividend_yield, debt_capital, market_beta, eps_growth)
drop if missing(dividend_yield, debt_capital, market_beta, eps_growth)
assert inrange(debt_capital, 0, 100)
sort industry
save "output/industry_data.dta", replace
export delimited using "output/industry_data.csv", replace
summarize dividend_yield debt_capital market_beta eps_growth

* 2. Linear, quadratic and cubic OLS; common industry sample.
generate double debt_sq = debt_capital^2
generate double debt_cube = debt_capital^3
tempname coef stats
postfile `coef' str12 model str20 term double coefficient se t p using ///
"output/coefficients.dta", replace
postfile `stats' str12 model double n r2 adj_r2 using "output/model_statistics.dta", replace
foreach model in linear quadratic cubic multiple {
    local rhs debt_capital
    if "`model'" == "quadratic" local rhs debt_capital debt_sq
    if "`model'" == "cubic" local rhs debt_capital debt_sq debt_cube
    if "`model'" == "multiple" local rhs debt_capital market_beta eps_growth
    regress dividend_yield `rhs'
    estimates store `model'
    post `stats' ("`model'") (e(N)) (e(r2)) (e(r2_a))
    foreach term in `rhs' _cons {
        post `coef' ("`model'") ("`term'") (_b[`term']) (_se[`term']) ///
            (_b[`term']/_se[`term']) (2*ttail(e(df_r),abs(_b[`term']/_se[`term'])))
    }
    if "`model'" != "multiple" predict double fit_`model', xb
    if "`model'" == "quadratic" test debt_sq
    if "`model'" == "cubic" {
        test debt_cube
        test debt_sq debt_cube
    }
}
postclose `coef'
postclose `stats'
tway (scatter dividend_yield debt_capital, mcolor(gs9) msize(small)) ///
(line fit_linear debt_capital, sort lcolor(navy)) ///
(line fit_quadratic debt_capital, sort lcolor(maroon) lpattern(dash)) ///
(line fit_cubic debt_capital, sort lcolor(forest_green) lpattern(dot)), ///
title("Dividend yield and debt-to-capital") ///

```

```

subtitle("U.S. industries | January 2026") ///
xtitle("Market debt-to-capital, lease-adjusted (%)") ytitle("Dividend yield (%)") ///
legend(order(1 "Industries" 2 "Linear" 3 "Quadratic" 4 "Cubic") rows(1) position(6)) ///
xlabel(0(20)80, format(%3.0f)) ylabel(0(2)8, format(%3.0f)) ///
graphregion(color(white)) name(assignment3_plot, replace)
graph export "output/dividend_debt_fits.png", width(1800) replace
graph save "output/dividend_debt_fits.gph", replace

```

```

* 3. Multiple linear regression: conventional OLS standard errors.
estimates restore multiple
estimates table multiple, b(%10.6f) se(%10.6f) t(%10.4f) stats(N r2)
preserve
use "output/coefficients.dta", clear
export delimited using "output/coefficients.csv", replace
restore
preserve
use "output/model_statistics.dta", clear
export delimited using "output/model_statistics.csv", replace
restore
save "output/industry_analysis.dta", replace
log close

```